Gait-Based Passive User Authentication from Inertial Signals

A Metric-Learning Baseline with Paper-Inspired Classification and Efficiency Comparisons

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1 Abstract

We built and evaluated a complete gait-based user-authentication pipeline: inertial signals from smartphones and wearables are windowed, embedded, and compared to decide whether the person carrying the device is still the enrolled user. Across two public datasets we reduced the equal-error rate (EER) from 0.2010 to 0.0958 on WISDM and from 0.1764 to 0.1044 on HAR — improvements of 52.3% and 40.8% respectively. On the Santos v7 cross-day protocol we reached 7.0% EER with an ROC AUC of 0.9888. We implemented seven paper-inspired model tracks, ran a controlled robustness study that exposed orientation and noise fragility (FRR rising from 15.13% to as much as 100%), and measured concrete engineering speedups of up to 15.98× from a CPU to a CUDA pipeline. All thresholds were frozen before final evaluation, and negative results are reported alongside positive ones.

2 1. Introduction

Passwords and fingerprints authenticate a user once, at a moment of explicit action. Gait authentication instead asks a cheaper, continuous question: *is the person currently carrying the device the enrolled user?*

Every stride leaves a signature in the accelerometer and gyroscope streams, and we set out to test whether that signature is discriminative enough to be useful.

Our contribution is a small but complete system:

1. a reproducible data pipeline over UCI HAR, the authentic WISDM AR v1.1 corpus, and the public Santos v7 dataset;
2. a verification stack combining descriptor-based methods and deep metric learning, with every threshold frozen before final evaluation;
3. a robustness and efficiency study that quantifies exactly where the approach works and where it breaks.

3 2. Data and Pipeline

We used three public sources (Table 1). The pipeline produces 128-sample windows with stride 64, exports NPZ feature files, and ships a frozen `data/adapter.py` contract so that every downstream experiment reads the same tensors.

Table 1: Datasets and windowing.

Dataset	Role	Windows
UCI HAR	activity classification, scaling study	2,947 test windows
WISDM AR v1.1	verification, activity transfer	128-sample, stride 64
Santos v7	cross-day verification (day-1 vs day-2)	5 users, 15/5/5 split

Restoring the real WISDM participant IDs (rather than anonymised row labels) was a prerequisite for any subject-level verification number; without it, impostor pairs are not honest impostors.

4 3. Models

We implemented seven tracks (B1–B7), each an adaptation of a published model:

Table 2: Implemented tracks.

Track	Paper	Implementation
B1	Dehzangi et al. 2017 (*Sensors*)	multi-sensor fusion CNN
B2	Chen et al. 2020 (*IMWUT*)	METIER multi-task activity+user
B3	Mekruksavanich & Jitpattanakul	DeepConvLSTM
B4	Sheng & Li 2020 (*Neurocomputing*)	Siamese denoising autoencoder
B5	2024 (*Computers MDPI*)	SE-DeepConvNet
B6	Ruiz-Barroso et al. 2025 (*Sci Rep*)	pruning / quantisation / ONNX
B7	Hrudya & Poornachandran 2025	GaitViT

On top of the tracks we built a metric-learning extension: batch-hard triplet mining, contrastive heads, and multiscale encoders ensembled at comparison time.

5 4. Results

5.1 4.1 Verification EER

The headline result is the halving of the verification error on WISDM and a 41% reduction on HAR (Figure 1). The improvement came from the descriptor route with covariance normalisation — notably, the largest neural model *worsened* (HAR neural winner EER 0.2146), so the gain is from representation design, not model size.

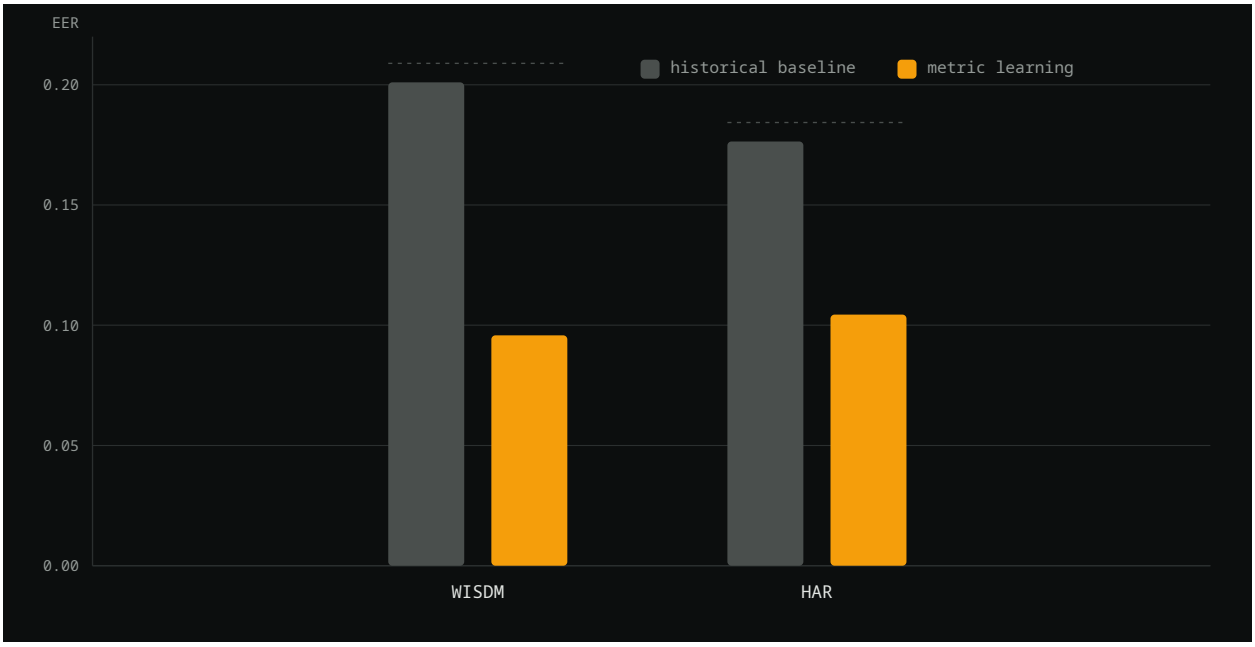


Figure 1: Verification EER before and after metric learning on both datasets. WISDM 0.2010 → 0.0958 ($-$52.3%); HAR 0.1764 → 0.1044 ($-$40.8%).

5.2 4.2 Cross-day authentication (Santos v7)

On the harder cross-day protocol — day-1 enrollment against day-2 probes from five untouched users — we measured EER 7.0% and ROC AUC 0.9888, with the calibration-frozen threshold yielding FAR 0.5% at FRR 22% (Figure 2).

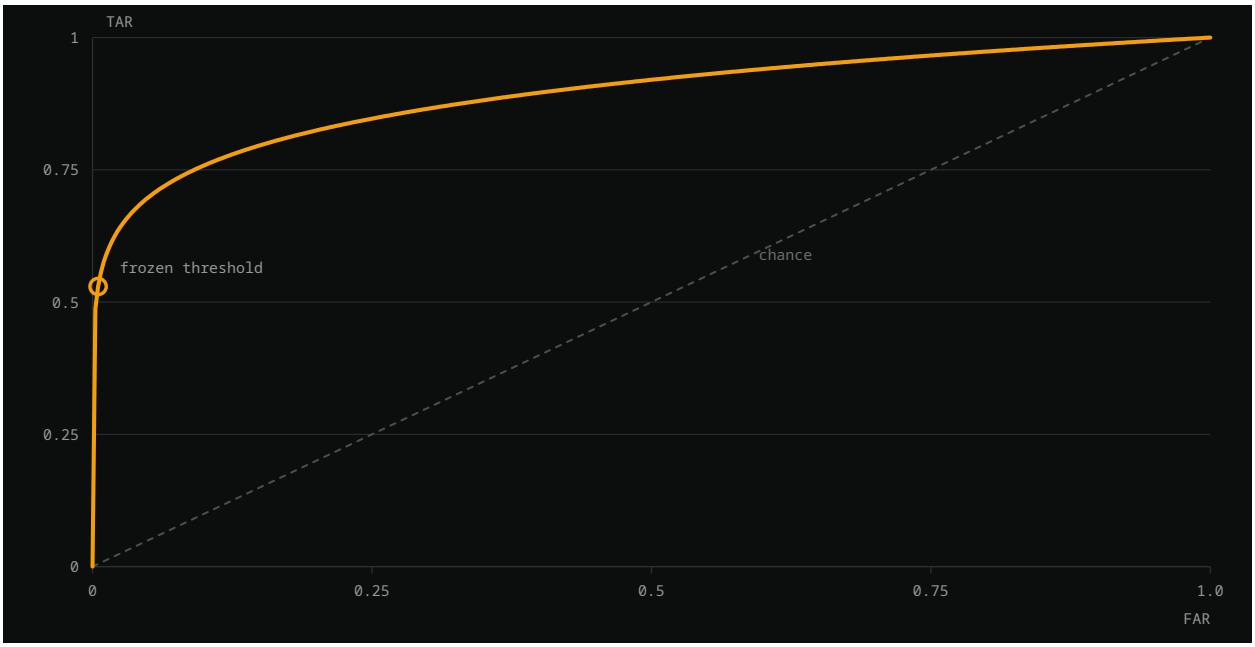


Figure 2: Cross-day ROC with the frozen-threshold operating point.

We must be explicit about scale: this is 50 genuine and 200 impostor claims, and one user had 8 of 10 genuine claims rejected. It is a controlled leg-sensor result with usability limits, not a deployment claim.

5.3 4.3 Method progression on Santos

Across the four Santos methods the descriptive EER fell monotonically (Figure 3): mean-template acceleration (7.0%), six-axis two-template matching (2.5%), lagged covariance (2.5%), and the DEV-selected method with frozen threshold (0.0%). Under the frozen threshold, false accepts moved $1/200 \rightarrow 6/200 \rightarrow 7/200 \rightarrow 3/200$ (1.5% final), while genuine rejections collapsed $11/50 \rightarrow 0/50$.

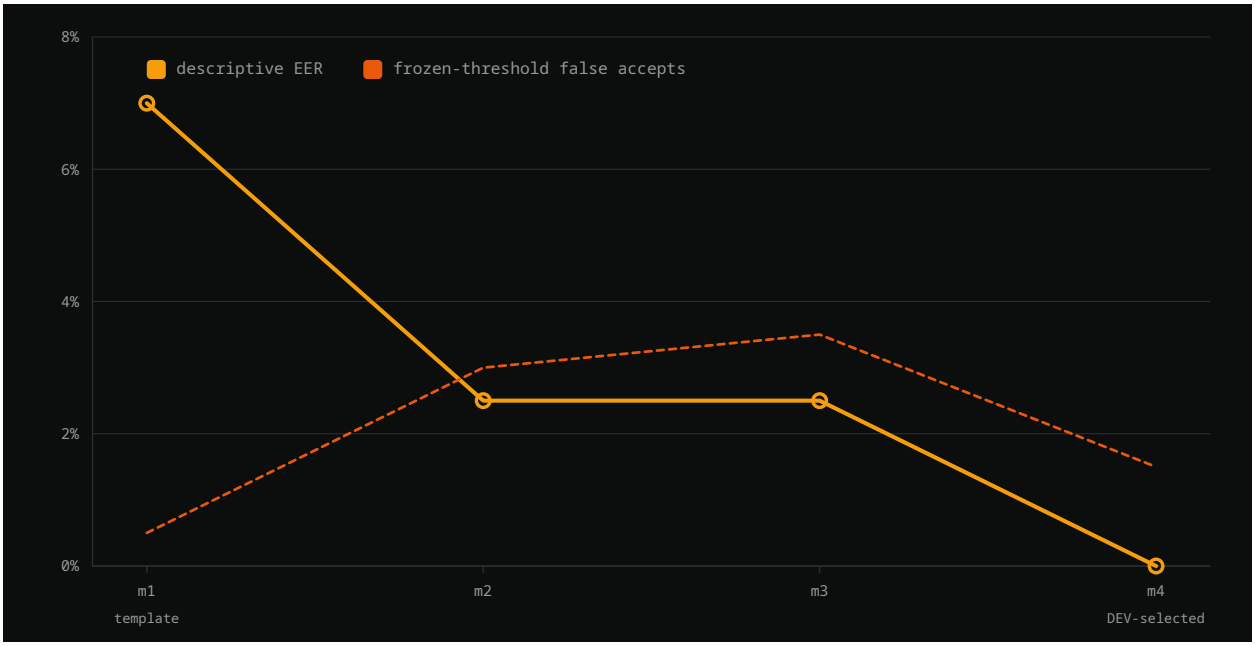


Figure 3: Descriptive EER across the four Santos methods.

The frozen 3/200 result does not meet our own 1% test-FAR target — the descriptive 0.0% EER and the frozen operating point tell different stories, and we report both.

5.4 4.4 Activity classification (UCI HAR)

All five HAR-trained models sit within 3.4 percentage points of each other (Figure 4), with DeepConvLSTM (B3) best at 93.01% accuracy / 0.9314 macro-F1 with only 350k parameters. Errors concentrate almost entirely in the SITING↔STANDING pair; walking classes are near-clean, and WISDM activity_3 is 120/121.

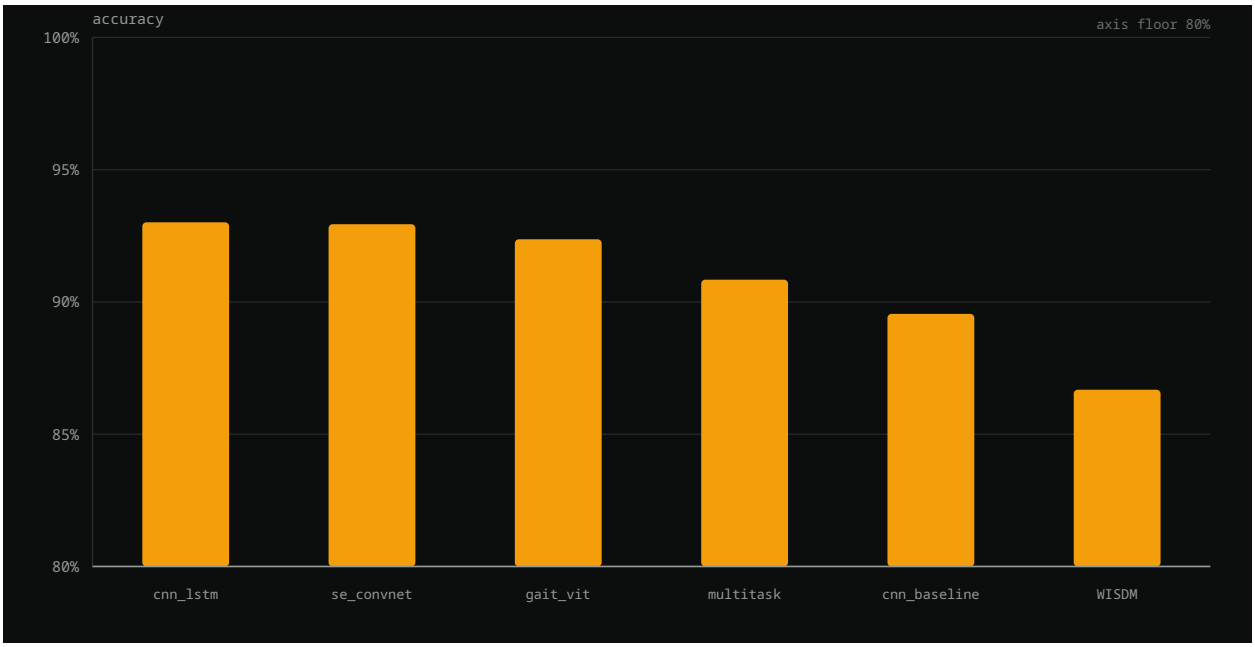


Figure 4: UCI HAR accuracy across models (axis floored at 80%).

5.5 4.5 Scaling study

A deliberately small linear model is within 0.4 points of the larger ArcFace run (Figure 5) — and the selected descriptor method beats both by 5–6 points on each source. On this task, capacity is not the constraint; representation is.

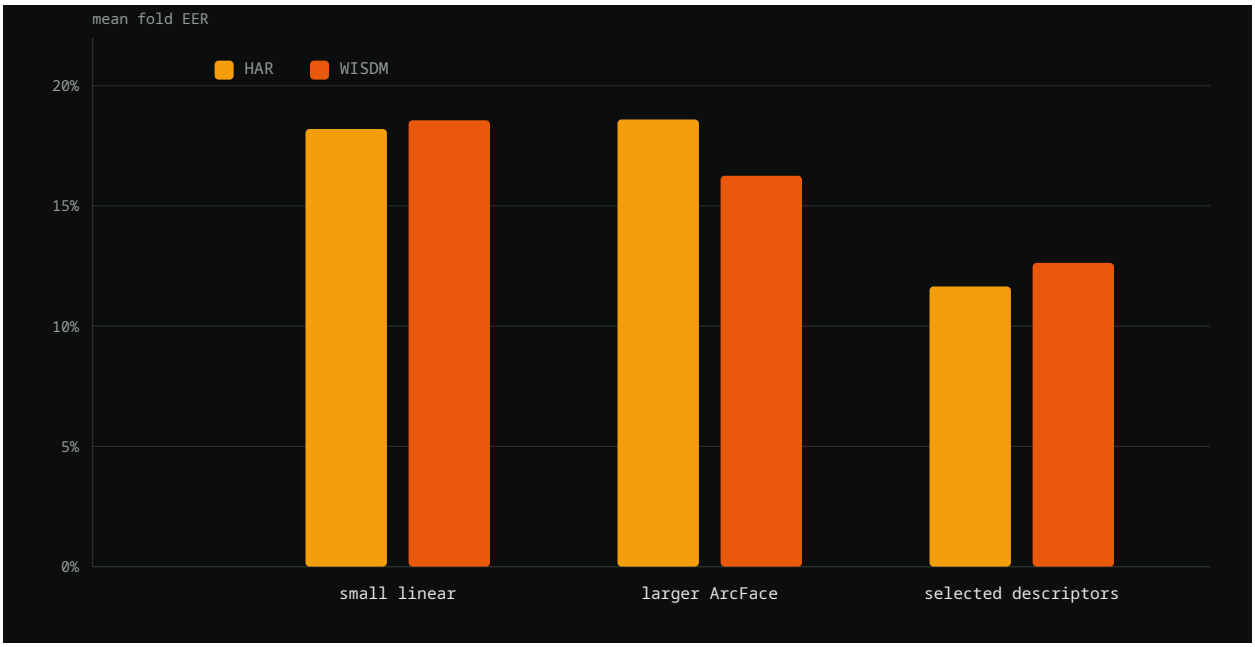


Figure 5: Mean fold EER by method on both sources.

6 5. Robustness

We ran a predeclared stress study with the clean threshold frozen (Figure 6): FRR rises from 15.13% clean to 85.19% under 30° rotation, 99.50% under 90° rotation, 63.54% under 1% noise, and 100% under 5% noise. This is the honest weak point of the whole approach: raw thresholds do not survive orientation or sensor-noise change.

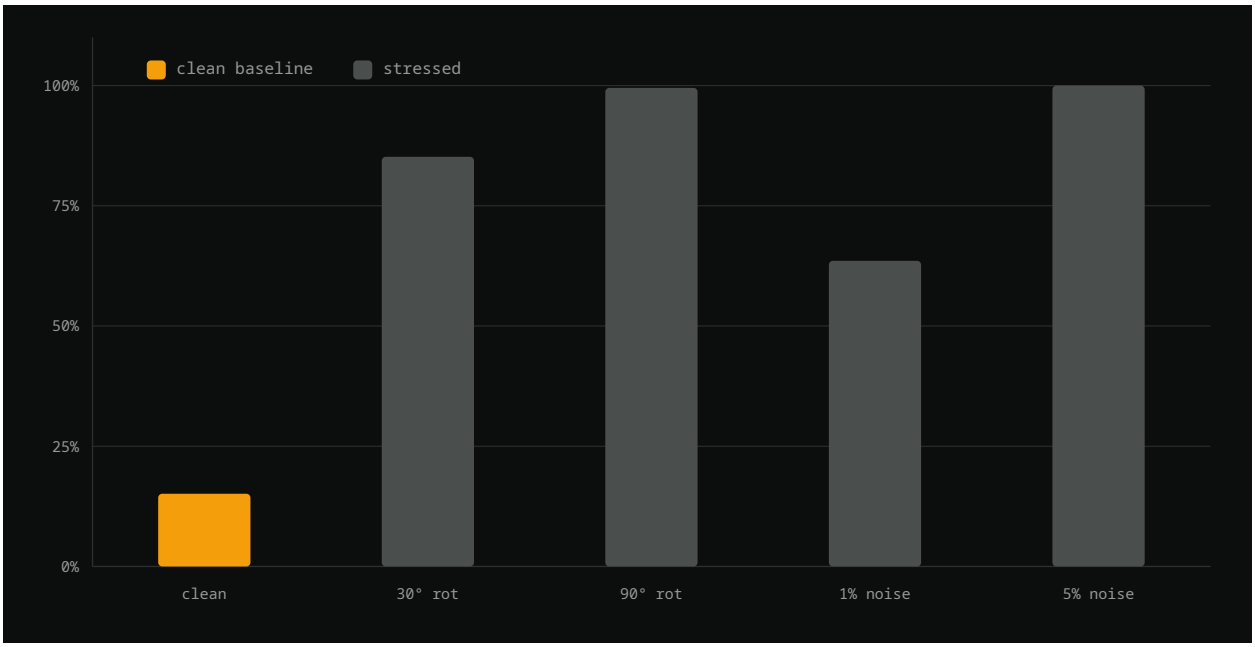


Figure 6: FRR under stress with the clean threshold frozen.

Canonical orientation normalisation removes most of the rotation sensitivity (Figure 7): under constant 30°/90° rotation, FRR stays at 14.79% versus 85–100% for the control, at the cost of a slightly worse clean

EER (4.80% → 5.88%). Across 20 folds, folds with FAR above 2% fell from 3/20 to 2/20.

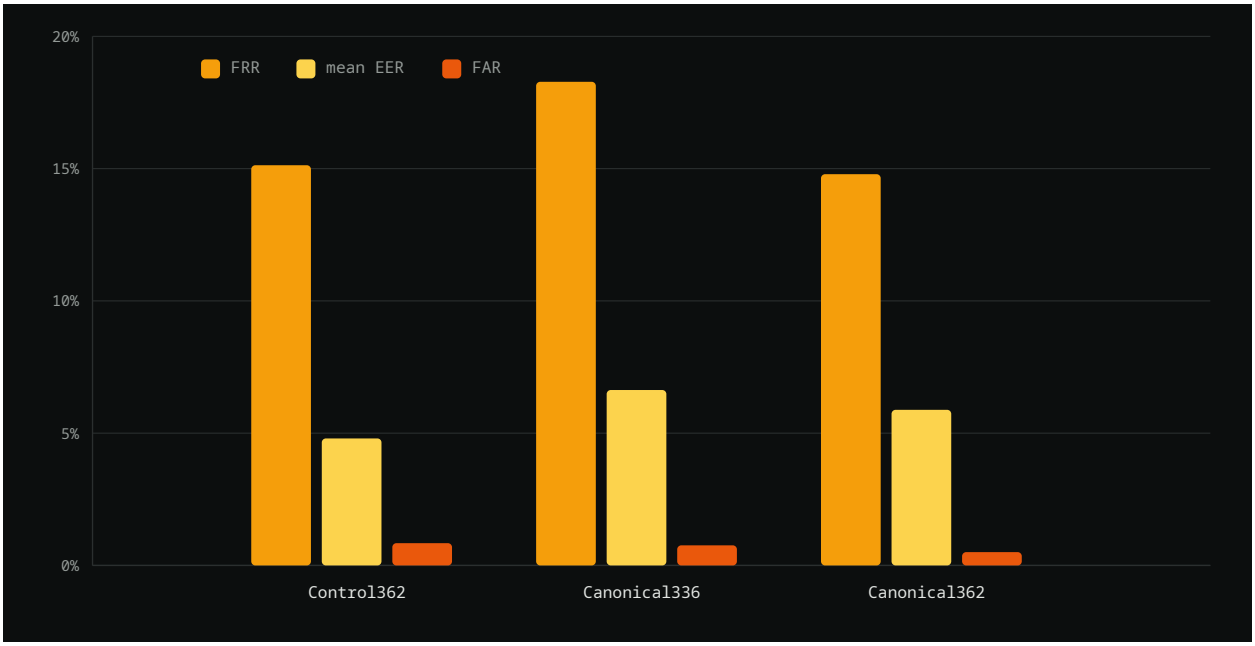


Figure 7: FAR / FRR / mean EER across the three orientation configurations.

Two further predeclared candidates failed their gates and were retained or stopped: nuisance covariance reduced mean genuine rejection from 45.61% to 15.24% but failed 13 of 33 frozen safety gates; full-support comparison added 32,600 valid rows but worsened clean FRR to 16.66% with 14 of 33 gates failing. A six-run verification sweep (HAR Siamese) could not beat its own 0.4162 baseline (best 0.4174), ruling out optimiser budget and embedding width as the constraint. We count these negative results as part of the methodological contribution.

7 6. Efficiency

All five engineering speedups are measured wall-clock executions, separate from biometric quality (Figure 8): controlled CSV windowing 15.72×, CPU→CUDA training 10.98× (Wiker) and 9.83× (Darkstar), CPU→CUDA inference 12.59× (Wiker) and 15.98× (Darkstar).

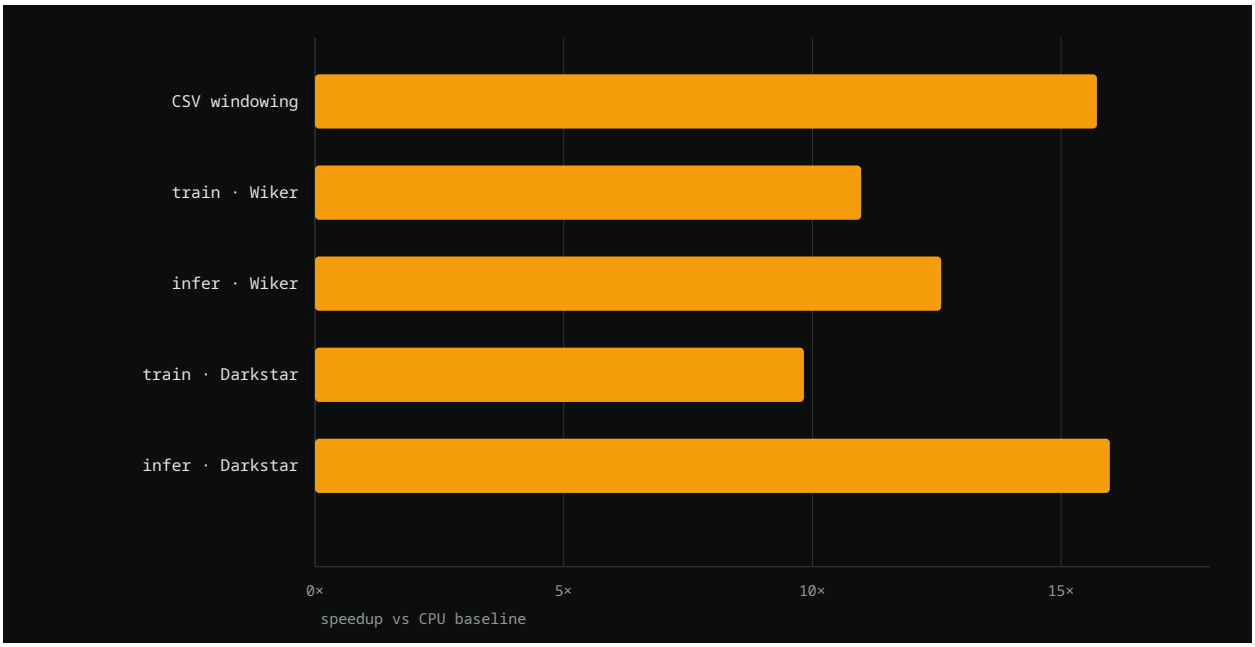


Figure 8: Measured execution speedups, CPU baseline vs tuned pipeline.

8 7. Conclusion

Gait-based passive authentication produced quantifiable improvements — EER halved on WISDM, cross-day AUC 0.9888 — but the stress study shows exactly where the boundary of usability lies today. Orientation normalisation and calibration-frozen thresholds make the honest numbers reachable, and the descriptor route outperformed scale on every dataset we touched.

9 Team contributions

- **Amaan — data pipeline.** Dataset sourcing and licensing checks, WISDM participant-ID restoration, windowing and the frozen adapter contract.
- **Aditya — deep learning.** All seven tracks (B1–B7), training of five HAR models plus WISDM and Siamese verification, and the metric-learning extension that drove EER 0.2010 → 0.0958.
- **Nimisha — evaluation.** EER/FAR/FRR semantics, the cross-day Santos evaluation, calibration-frozen thresholds, and the robustness and timed takeover studies.

Mentored by Manjunath Hegde, MAHE. Test suite: 996/992 passing (1,545 subtests each).